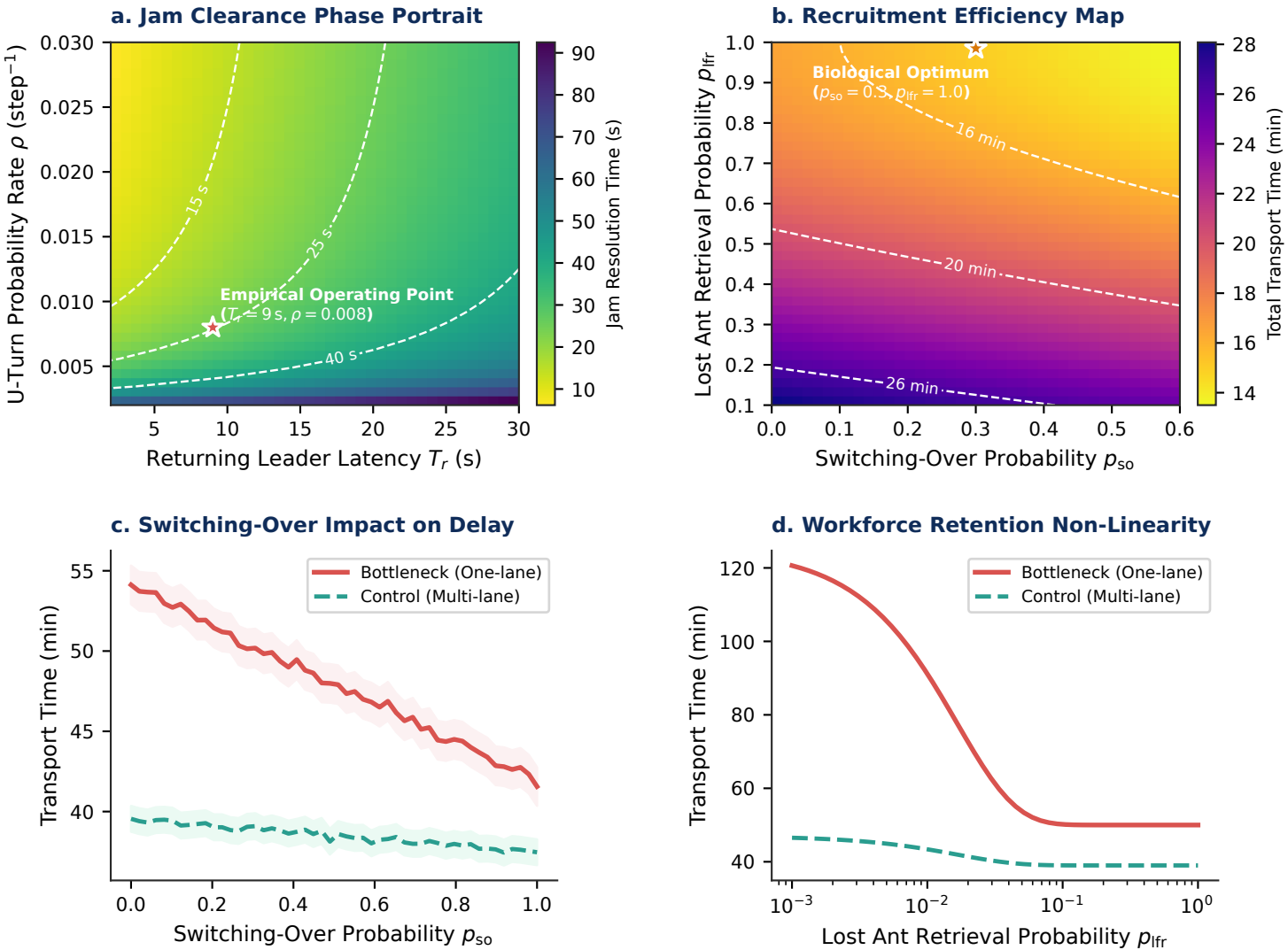
a. Jam Clearance Phase Portrait
U-Turn Probability Rate ρ (step⁻¹)
Returning Leader Latency T_r (s)
Jam Resolution Time (s)
15 s
25 s
40 s
Empirical Operating Point
($T_r = 9$ s, $\rho = 0.008$)
b. Recruitment Efficiency Map
Lost Ant Retrieval Probability p_{lfr}
Switching-Over Probability p_{so}
Total Transport Time (min)
Biological Optimum
($p_{so} = 0.3$, $p_{lfr} = 1.0$)
16 min
20 min
26 min
c. Switching-Over Impact on Delay
Transport Time (min)
Switching-Over Probability p_{so}
Bottleneck (One-lane)
Control (Multi-lane)
d. Workforce Retention Non-Linearity
Transport Time (min)
Lost Ant Retrieval Probability p_{lfr}
Bottleneck (One-lane)
Control (Multi-lane)